

HOMEWORK 5

1. Let F be a perfect field and $f \in F[x]$ a nonzero polynomial. Prove that the factor ring $F[x]/fF[x]$ is semisimple if and only if f is separable.
2. Prove that an infinite product of fields is not a semisimple ring.
3. Let R be a ring such that all (left) cyclic R -modules are projective. Prove that R is semisimple.
4. Let R be a ring such that every two simple (left) R -modules are isomorphic. Prove that R has no nontrivial central idempotents.
5. Find a decomposition of the monomial representation $S_n \rightarrow GL_n(\mathbb{Q})$ into a direct sum of irreducible representations.
6. Prove that $\mathbb{Q}[S_3] \simeq \mathbb{Q} \times \mathbb{Q} \times M_2(\mathbb{Q})$.
7. Let G be the quaternion group. Prove that $\mathbb{R}[G] \simeq \mathbb{R} \times \mathbb{R} \times \mathbb{H}$, where $\mathbb{H}$ is the quaternion algebra.
8. Determine all irreducible representations of $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Q}$.
9. A representation $G \rightarrow GL_n(F)$ over a field F is called *absolutely irreducible* if the composition $G \rightarrow GL_n(F) \rightarrow GL_n(\overline{F})$, where $\overline{F}$ is an algebraic closure of F , is irreducible. Prove that if $G \rightarrow GL_n(F)$ is absolutely irreducible, then for every field extension L/F , the composition $G \rightarrow GL_n(F) \rightarrow GL_n(L)$ is irreducible.
10. Let G be a finite group, $H \subset G$ a subgroup, V an H -space. Consider the vector space W of all maps $f : G \rightarrow V$ satisfying $f(hg) = hf(g)$ for all $h \in H$ and $g \in G$. Show that the G -action on W given by $(gf)(g') = f(g'g)$ for all $g, g' \in G$ and $f \in W$ makes W a G -space.